

AWS Certified Cloud Practitioner (CLF-C02)

Exam Essentials Guide

By Chetan Agrawal (www.awswithchetan.com)



Exam Essentials Guide – Preface

- This guide is a concise revision companion designed to reinforce the most important exam-relevant concepts across AWS Cloud and Services. It summarizes key ideas, behaviors, limits, and best practices exactly as covered in the course lectures, helping you quickly recall what matters most during the exam.
- The content is not a replacement for lectures or hands-on practice, but a high-impact reference for final revision, concept clarity, and exam readiness. Each slide focuses on frequently tested patterns, common traps, and decision-making cues that appear in AWS certification exams.
- Use this guide for last-mile preparation, quick refreshers, and confidence building before the exam.



Exam domains

- Domain 1: Cloud Concepts (24% of scored content)
- Domain 2: Security and Compliance (30% of scored content)
- Domain 3: Cloud Technology and Services (34% of scored content)
- Domain 4: Billing, Pricing, and Support (12% of scored content)



Cloud Concepts

- Cloud computing is the on-demand delivery of IT resources over the internet with pay-as-you-go pricing
- There are five characteristics of the cloud – On-demand & self-service, Broad network access, Multi-tenancy & resource pooling, Rapid elasticity & scalability and Measured services
- There are six advantages of the cloud – Pay-as-you-go, Economy of scale, Stop guessing capacity, Increased speed & agility, Stop paying for running and maintaining data centers and Go global in minutes
- Problems which are difficult to solve without the cloud – Scalability, Elasticity, High Availability & Fault tolerance and Flexibility
- Deployment models of the cloud – Private Cloud, Public Cloud and Hybrid Cloud
- Types of Cloud computing – On-premises, Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS)



AWS Global Infrastructure

Amazon Region & AZs:

- Amazon Regions consist of multiple Availability Zones (AZs) located in distinct geographic areas, offering redundancy and isolation from failures in other regions.
- Choose region based on end user latency, data residency, DR requirements, AWS Services availability & pricing

AWS Edge Locations / CloudFront:

- Enables low-latency connectivity to AWS region through AWS backbone network.
- Can cache static content (e.g. video, images)
- **Can not** be used for hosting applications unlike Region, LZ, WLZ or Outpost.



AWS Global Infrastructure

AWS Local Zone:

- AWS Local Zones extend AWS infrastructure to specific geographic locations, enabling low-latency access to AWS services for applications that require single-digit millisecond latency.
- Choose AWS Local Zones when you need to place resources closer to end-users in **metropolitan** areas meeting low latency and compliance requirements.

AWS Wavelength:

- Extends AWS infrastructure to **5G mobile network** (Limited AWS services)
- Choose AWS Wavelength when you want to deploy ultra-low latency applications at the edge of 5G networks (e.g. Gaming, Connected vehicles)

AWS Outpost:

- Run AWS services **on-premises** (Limited AWS services)
- Use AWS Outposts when you need to run AWS infrastructure on-premises or in co-location facilities for low latency and local data processing.



AWS Identity and Access Manager (IAM)

- Root user - Owner of the AWS account. Root user has unrestricted permissions to AWS account. Do not use unless really required to.
- IAM User - A regular user who can have restricted access to AWS account.
- Use MFA + Password Policy for securing AWS account console login for root and IAM users
- IAM User Group - Group of IAM users. Can attach IAM policies to the group.
- IAM Policy - JSON document that outlines permissions for IAM users or groups or role.
- IAM Role - Provides temporary IAM credentials for AWS services (e.g. EC2, Lambda) or cross-account IAM users or federated users.
- Access Keys – For programmatically accessing AWS account using the CLI or SDK. Consists of Access key ID + Secret access key. Must keep Access & secret keys secure.
- AWS CLI - Manage your AWS services using the command-line interface
- AWS SDK - Manage your AWS services using a programming language (python, java, node, C# and more)
- AWS CloudShell – A browser based pre-authenticated shell to access AWS CLI from AWS console
- IAM Audit and governance - IAM Credential Reports, IAM Access Analyzer and IAM access advisor



Amazon EC2

- EC2 - Virtual Machine in the AWS cloud hosted in an AZ within the Region
- EC2 Configuration options - Instance Type/Size (CPU + RAM) + AMI (OS) + Storage (EBS/Instance store) + VPC + Security Groups + SSH key-pair (+ IAM Role, User data, Tags)
- EC2 Block Storage options – Persistent storage - Elastic Block Store (EBS), Non persistent storage - Instance Store
- EC2 Security groups – Contains inbound rules and outbound rules (Protocol, Port, IP range), Stateful
- EC2 initialization – EC2 User Data scripts
- Login Options: SSH, RDP (there are more options to connect to EC2 & we will cover them later)
- Purchasing Options - On-Demand, Spot, Reserved, Savings Plan
- Capacity Reservation (without commitment) - On-demand capacity reservation (ODCR)
- Tenancy options – Shared, Dedicated Instances, Dedicated Host
- Tags – Name-value key pairs, useful for filtering, deployment, User IAM permissions, cost allocation etc.



Other AWS Compute services

- Containers – Lightweight as compared to VMs. Repeatable, Consistent and Portable.
- ECS – Elastic Container service built by AWS. Runs docker containers at scale. Simplifies container management.
- ECS supports EC2 and Fargate as a compute platform for hosting containers.
- Fargate – Serverless compute option for running containers where infrastructure is fully managed by AWS.
- ECR – Elastic Container registry to store container images. Can be used by ECS or EKS.
- EKS – Elastic Kubernetes Service. A managed kubernetes cluster.
- Lambda – Serverless compute for event driven applications.
- Lambda is ideal for stateless executions which needs to run for short amount of time (& vanish) e.g. API backend, Event processing, Data processing etc.



Load Balancer and Auto Scaling Group

- High Availability in AWS means deploying application across multiple Availability Zones (AZs)
- For EC2 based applications high availability and scalability, use Elastic Load Balancer and Auto Scaling group

Elastic Load Balancer (ELB)

- For distributing incoming traffic across backend EC2 instances, containers, IP addresses or lambda function
- AWS has 4 types of load balancers: Application Load Balancer (Layer 7), Network Load balancer (Layer 4), Gateway Load Balancer (Layer 3), Classic Load Balancer (Layer 4 & 7)
- Load balancer performs health check of backend targets and send traffic to only healthy target

Auto Scaling group

- Scales the EC2 capacity up or down as per the auto scaling group policy
- Supports Manual scaling, Dynamic scaling (Simple, Target tracking), Scheduled scaling, Predictive scaling
- Works closely with ELB for dynamic scaling
- Provides cost benefits by running only required number of EC2 instances for applications serving unpredictable traffic/load



VPC and Networking

- VPC is a logically isolated private network in AWS
- VPC has a private address range called CIDR (IPv4 and IPv6)
- VPC supports Public/Elastic IPs for public connectivity and Private IPs for private connectivity.
- Internet Gateway connects VPC to the internet
- VPC has subnets – Public subnet and Private Subnet. Public subnet route table has route to go to internet via IGW.
- NAT gateway provides outbound internet access to instances in the Private subnet
- VPC offers 2 types of firewalls – Security group and Network ACL
- AWS creates default VPC in each region
- VPC endpoint provides private connectivity to other AWS services from within the VPC
- VPC peering connects two VPCs, non-transitive, non-overlapping addresses
- AWS Transit Gateway connects thousands of VPCs, On-premises network
- AWS Site-to-Site VPN for encrypted traffic from on-premises to AWS. Traffic goes over the internet.
- AWS Client VPN provides for encrypted traffic from your workstation to AWS VPC
- AWS Direct connect for physical connection from on-premises to AWS



DNS and Edge Services

Route 53 (DNS)

- Amazon Route 53 is Global DNS service and Domain Registrar service of AWS
- Create Public hosted zone for Public DNS and Private hosted zone for Private DNS within the VPC
- Route53 supports different DNS records e.g. A (IPv4), AAAA (IPv6), CNAME, Alias and more
- There are different routing policies e.g. Simple, Failover, Weighted, Geolocation, Latency based.

Edge Locations

- AWS has hundred plus edge locations across 90+ cities
- These edge locations are used by Amazon CloudFront, AWS Global Accelerator and AWS S3 Transfer accelerator to optimize network path from end user to the AWS region where applications are hosted.

Amazon CloudFront

- Amazon CloudFront is a Content Delivery Network (CDN) service of AWS
- Amazon CloudFront uses AWS Edge locations to optimize network path and cache the static contents e.g. static html files, images, videos etc.



DNS and Edge Services

AWS Global Accelerator

- AWS Global Accelerator provides 2 static IPs which clients can use to connect to Global accelerator endpoint
- AWS Global Accelerator supports Application Load Balancer, Network Load Balancer and EC2

AWS S3 Transfer Acceleration

- AWS S3 Transfer Acceleration (S3TA) is a feature of S3 bucket which enables edge locations to optimize network path and uses network optimization protocols to speed up the data transfer to s3. Ideal for large file uploads to s3.



Storage - Elastic Block Store (EBS)

- EBS is network drive providing block storage to the EC2 instance
- Generally, EBS volume is attached to a single EC2 instance but depending on EC2 instance types (based on nitro system), specific EBS volumes can be multi-attach.
- EBS volume types – General purpose (gp2, gp3), Provisioned IOPS (io1, io2), Throughput optimized HDD (st1), Cold HDD (sc1)
- EBS volumes are in single AZ
- EBS volumes can be backed up using EBS snapshot
- EBS snapshots are incremental. Full snapshot + incremental changes.
- For moving EBS volume across AZ, create a snapshot and then create a volume in other AZ.
- EBS volume has an attribute called 'DeleteOnTermination'. By default, for Root volume this flag is True and for non-root volumes it is False
- For long term retention of EBS data, create a snapshot and delete the volume.



Storage - File Systems

Elastic File System (EFS)

- A managed Network File System
- Can be shared by hundreds of EC2 Linux instances
- Works with most of the AWS compute services
- Serverless - no need to manage infrastructure
- Elastic – no need to pre-provision the capacity, pay as per the storage used
- Supports different storage classes like EFS Standard, Standard Infrequent Access (IA) and EFS Archive

FSx for Windows File Server

- A fully managed, reliable and scalable Windows shared file system
- Uses Windows File Server
- Supports Windows NTFS and SMB protocols

FSx for Lustre

- High-performance File System for HPC workloads shared by thousands of compute machines.
- Data stored in Amazon S3 is loaded to FSx for Lustre for processing.

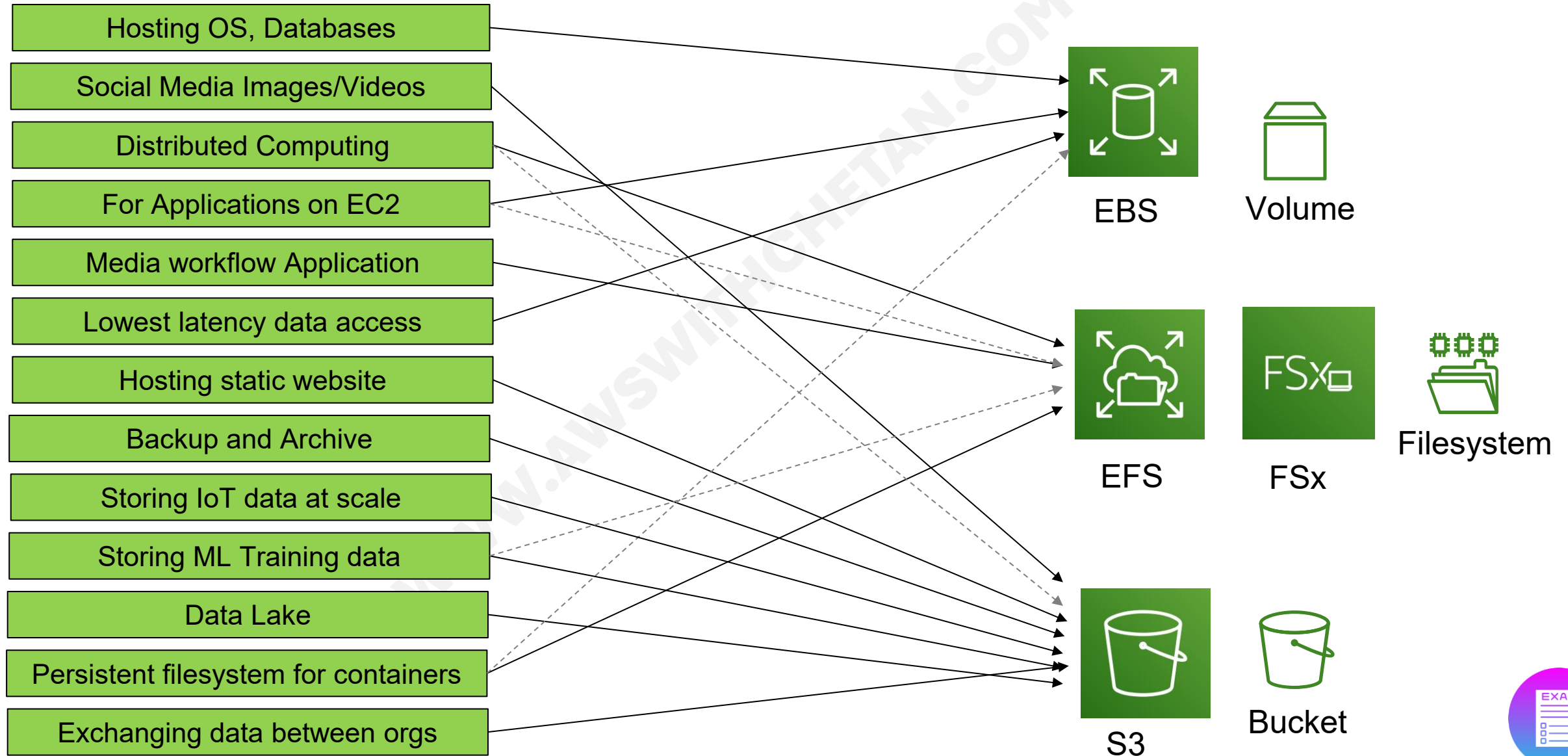


Storage - Amazon S3

- S3 is an object storage which is accessible directly over the web (http/https)
- S3 bucket contains Objects (max size 50TB), unlimited number of objects in the bucket.
- S3 supports different storage classes for different storage requirements based on latency, frequency of access and durability of the data.
- S3 Storage classes: S3 Standard, S3 Standard-IA, S3 One Zone-IA, S3 Intelligent-Tiering, S3-Glacier*
- S3 Security - Block Public Access, IAM user policy, Bucket Policy, Bucket/Object ACL,
- S3 Data encryption using Server-side encryption (SSE-S3, SSE-KMS, SSE-C) and Client-side encryption
- S3 lifecycle configurations to transition or expire objects after specific duration
- S3 can host a static website (not required to have server-side processing e.g. PHP, python)
- S3 supports versioning where multiple versions for objects are stored. Helps to recover objects in case of accidental deletion.
- S3 Replication - SRR (same-region) or CRR (cross-region). For replication versioning must be enabled.
- IAM access analyzer can identify public s3 buckets and buckets accessible by other AWS accounts

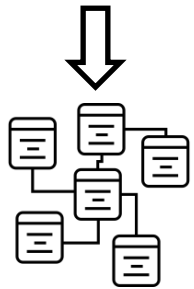


AWS Storage services - When to use what?



AWS Database services

Banking,
Finance,
Bookings,
ERP, CRM

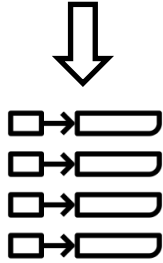


Relational
Database



Amazon RDS Amazon Aurora

Shopping cart,
Product catalog,
Customer attributes



Key-value or
NoSQL
Database



Amazon
DynamoDB

Content
management,
personalization,
mobile



Document
Database



Amazon
DocumentDB

Leaderboards,
real-time
analytics,
caching

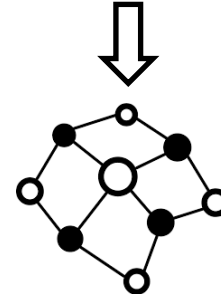


In-memory
Database



Amazon
ElastiCache

Fraud detection,
social
networking,
recommendation engine



Graph
Database



Amazon
Neptune

IoT
applications,
event tracking

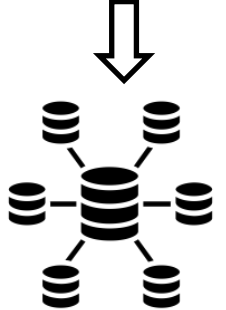


Timeseries
Database



Amazon
Timestream

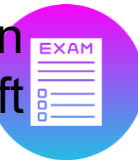
Analytics,
Data Marts



Data
Warehouse



Amazon
Redshift



Amazon RDS

- A managed Relational Database service where AWS handles provisioning, patching, upgrades, backup, recovery, repair, monitoring etc.
- RDS supports Automatics continuous backups and on-demand or scheduled snapshots.
- You do not get access to underlying EC2 instances which host RDS

RDS Read Replicas

- Supports different database engines like - MySQL, Postgres, MariaDB, Oracle, SQL Server
- Supports RDS Read-replicas - To scale the read queries thereby relieving pressure on Primary DB
- In case of failure of the Primary DB, read replica can be promoted to standalone DB

RDS Multi-AZ

- Primary DB instance in one AZ and Standby DB instance(s) in another AZ.
- Application reads from and writes to only Primary DB.
- Data is synchronously replicated from Primary DB to Standby DB.



Amazon Aurora

- Aurora is PostgreSQL and MySQL compatible relational database engines built by AWS.
- Aurora claims to provide 5x the throughput of MySQL and 3x of PostgreSQL DB and 1/10th cost of commercial databases.
- Aurora maintains 6 copies of data across 3 Availability Zones.
- Aurora supports massive read scaling with up to 15 read replicas.
- Amazon Aurora key features:
 - Aurora Global Database - Data in Primary DB (in the primary region) is replicated to other regions (within 1 sec)
 - Aurora Serverless - Automatically starts up, shuts down, and scales capacity up or down based on your application's needs



Amazon DynamoDB

- Amazon DynamoDB is a fully managed, serverless cloud-native **NoSQL** database service
- Provides millisecond latency at any scale
- Has Partition key, Sort key (optional) and attributes. Partition key + Sort key = Primary Key
- Support Query and Scan operations – Query fetches specific item; Scan fetches all items in the table.
- Supports Global tables (active-active) for global applications.
- Support DynamoDB accelerator (DAX) providing microsecond latency
- Supports DynamoDB streams to captures the item level modifications in time-ordered sequence. Can be used to act on table level modifications (e.g. using Lambda function as subscribers to the stream).



Other Databases

Amazon DocumentDB

- Used to store, query, and index JSON-like documents like User profile, blogs, video metadata, eCommerce product catalog etc.

Amazon ElastiCache

- In-memory databases with high performance and extremely low latency to read/write data.
- Helps reduce load off database I/O for read intensive workloads.
- Serverless, Highly available, Multi-AZ with cross region replication
- Supports Redis OSS, Memcached and Valkey engines. Use Memcached for simple use cases and Redis for advanced.

Amazon Neptune

- Graph databases store data as a network of entities and relationships.
- Can perform similarity searches on vectors stored along with your graph for gen AI apps.

Amazon Timestream

- Time series Database used for Live analytics, IoT data ingestion and analytics, Operational metrics.

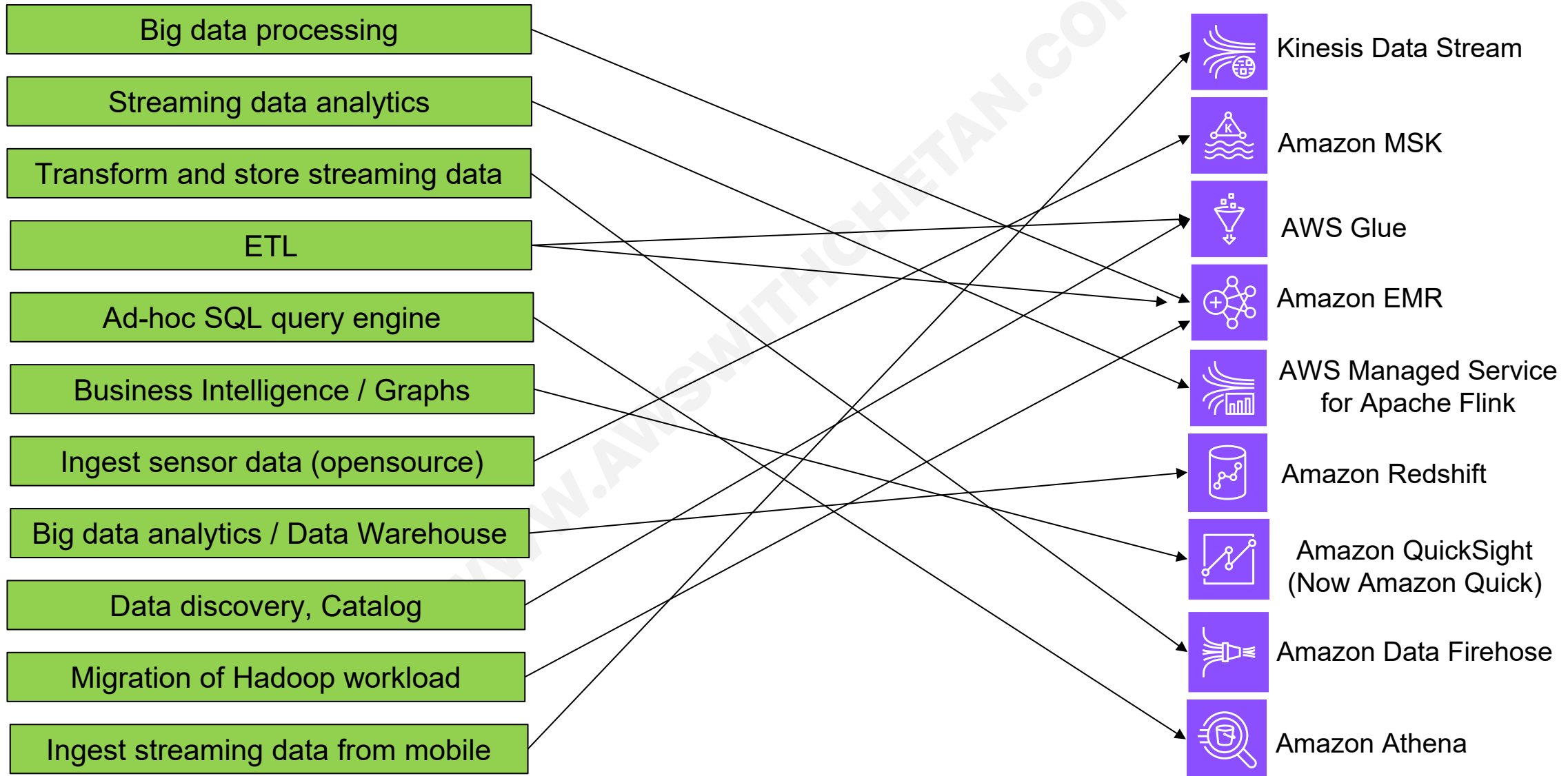


AWS Analytics Services

- OLTP – Online Transaction Processing – Amazon RDS, Aurora and NoSQL databases
- OLAP – Online Analytical Processing – Amazon Redshift
- Data Warehouse - System that stores and organizes data from multiple sources for reporting and analysis (think Amazon Redshift)
- Data Lake – Centralized repository to store large amount of data (think Amazon S3)
- Amazon Kinesis Data Streams – Real-time data ingestion (Connected cars, IoT, machine sensor, click stream etc.)
- Amazon Kinesis Video Streams – Real-time video ingestion (Smart cameras, Home security, Driver behavior etc.)
- Amazon Managed Service for Apache Flink – Real-time data analysis (Fraud detection, Ad, recommendations etc.)
- Amazon Data Firehose – Store streaming data into Amazon S3, Redshift or OpenSearch
- AWS Glue – Managed ETL service with Schema discovery and Data Cataloging
- Amazon EMR – Hadoop platform to run big data frameworks for ETL and big data processing
- Amazon Athena – Interactive, serverless SQL query engine to query data stored in S3 and other data sources
- Amazon Redshift – Petabyte scale Data warehouse service (similar to Snowflake, Google BigQuery)
- Amazon QuickSight (Now Amazon Quick) – Machine learning powered Business Intelligence (BI) service



AWS Analytics services - When to use what?



AWS Application Services

Amazon API gateway

- RESTful and HTTPS APIs (application programming interfaces)
- Supports features like caching, authentication, authorization, API keys, Throttling, Custom domain name etc.

AWS AppSync

- Serverless and high-performing GraphQL and Pub/Sub API service
- Based on a predefined GraphQL schema that defines the data structure and operations.
- Connects to DynamoDB, Aurora, Elasticsearch, HTTP endpoints, and Lambda functions.

Amazon SQS

- Queue service in AWS, supports Standard Queue and FIFO queue
- Used to decouple applications in AWS.
- Messages can be retained for up to 14 days

Amazon SNS

- Notification service in AWS
- Contains Topics and Subscribers: Email, Lambda, SQS, HTTP, Mobile phone numbers (SMS), Push notifications



AWS Application Services

Amazon EventBridge

- Event bus for event-driven applications
- Allow to connect applications based on the events
- Event filters and rules to process and event before sending it to the target

Amazon MQ

- Managed message broker for ActiveMQ and RabbitMQ in the cloud



Infrastructure and Application deployment

Infrastructure As a Code (IaC)

- AWS CloudFormation – Infrastructure as a code (YAML or JSON templates)
- CDK – Cloud Development Kit to create, share, deploy CloudFormation template programmatically

DevOps CI/CD services

- CodeCommit – Git compatible source code depository
- CodeBuild – Compile source code and run tests
- CodeDeploy – Deploy the code builds onto the compute servers (EC2, on-premises, ECS, Lambda)
- CodePipeline – End to end CI/CD pipeline automation
- CodeArtifact – Software repository / package management for software development and build

Elastic Beanstalk

- Fully managed platform-as-a-service (PaaS) for 3-tier web applications (ALB, ASG, EC2).
- Developer's friendly where no-infrastructure management needs to be done.

Amazon Lightsail

- Preconfigured servers with fix capacity and price for simple applications, databases and containers.



Infrastructure Management and Compliance

AWS Systems Manager (SSM)

- Automates common and repetitive IT operations and management tasks
- Centrally manage EC2 instances and on-premises servers
- Works for Linux, Windows, MacOS, and Raspberry Pi OS (Raspbian)
- Run Command - Author and execute runbooks across fleet of EC2 or on-premises servers
- Fleet Manager - Get operational insights about the state of your infrastructure
- Patch Manager - Select and install OS and software security patches
- Session Manager - Allows you to start a SSH session to your EC2 instance and on-premises servers.

AWS Config

- Record and track AWS resources configuration changes
- Run compliance checks against ideal configurations and take remediation action
- On detecting any configuration deviation - Send SNS notification or invoke API call to remediate the changes.



Infrastructure Monitoring and Logging

Amazon CloudWatch

- Collects and tracks metrics, logs, and events for AWS resources and applications.
- Create custom dashboards for a visual representation of metrics and logs, aiding in monitoring and decision-making.
- Supports monitoring for most of the AWS services (e.g., EC2, RDS, Lambda) and third-party apps via API.
- CloudWatch Metric - The unit of measurement for system performance e.g. CPU Utilization, Network In/Out, Disk I/O
- CloudWatch Alarm - Automatically trigger actions (e.g., scaling, notifications) based on defined metrics thresholds.
- CloudWatch Logs
 - Collect, store, monitor and analyse logs from AWS services and applications such as EC2, ECS, Elastic Beanstalk, CloudTrail, Route53 DNS
 - Need to install CloudWatch Agent to push logs to CloudWatch. Agent needs IAM permissions to push the logs.
 - Logs can be stored in Standard storage class or Infrequent-Access (Logs-IA) at lower cost.

Note: EC2 basic monitoring publishes metrics every 5-minutes whereas detailed monitoring publishes metrics at every 1-minute



Infrastructure Monitoring and Logging

Amazon CloudTrail

- Get an history of events / API calls made within your AWS Account
- AWS CloudTrail help you answer the question of "Who did what and when in your AWS account?"
- A trail can be applied to All AWS Regions (default) or a single Region
- We can send logs from CloudTrail into CloudWatch Logs or Amazon S3 and then query using CloudWatch logs insights or Amazon Athena

AWS X-Ray

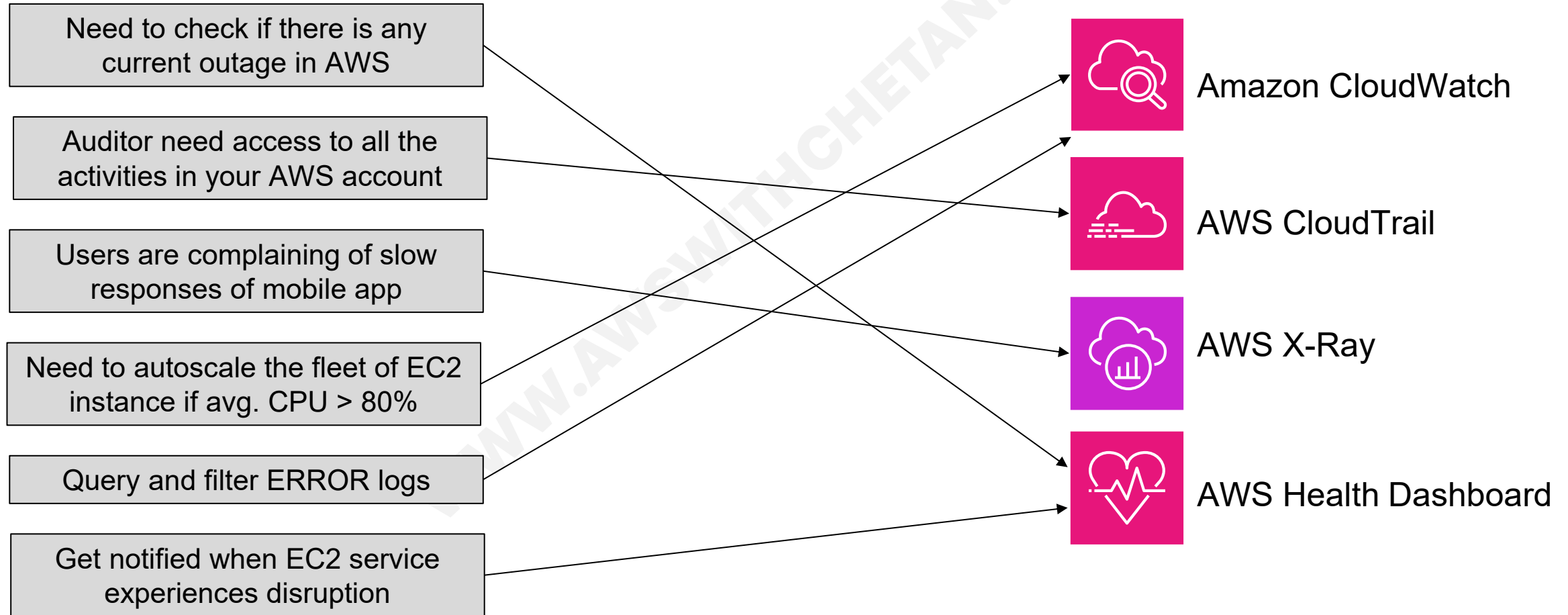
- Service that collects data for requests that your application serves.
- Debugging slow responses to the user requests or troubleshooting increased error rates
- Use cases: Which service is the bottleneck in the application? Which users are impacted by the service disruption?

AWS Health Dashboard

- Status of all AWS services across all regions
- AWS events having impact on your AWS services or resource in your AWS account



AWS Monitoring & Logging services - When to use what?

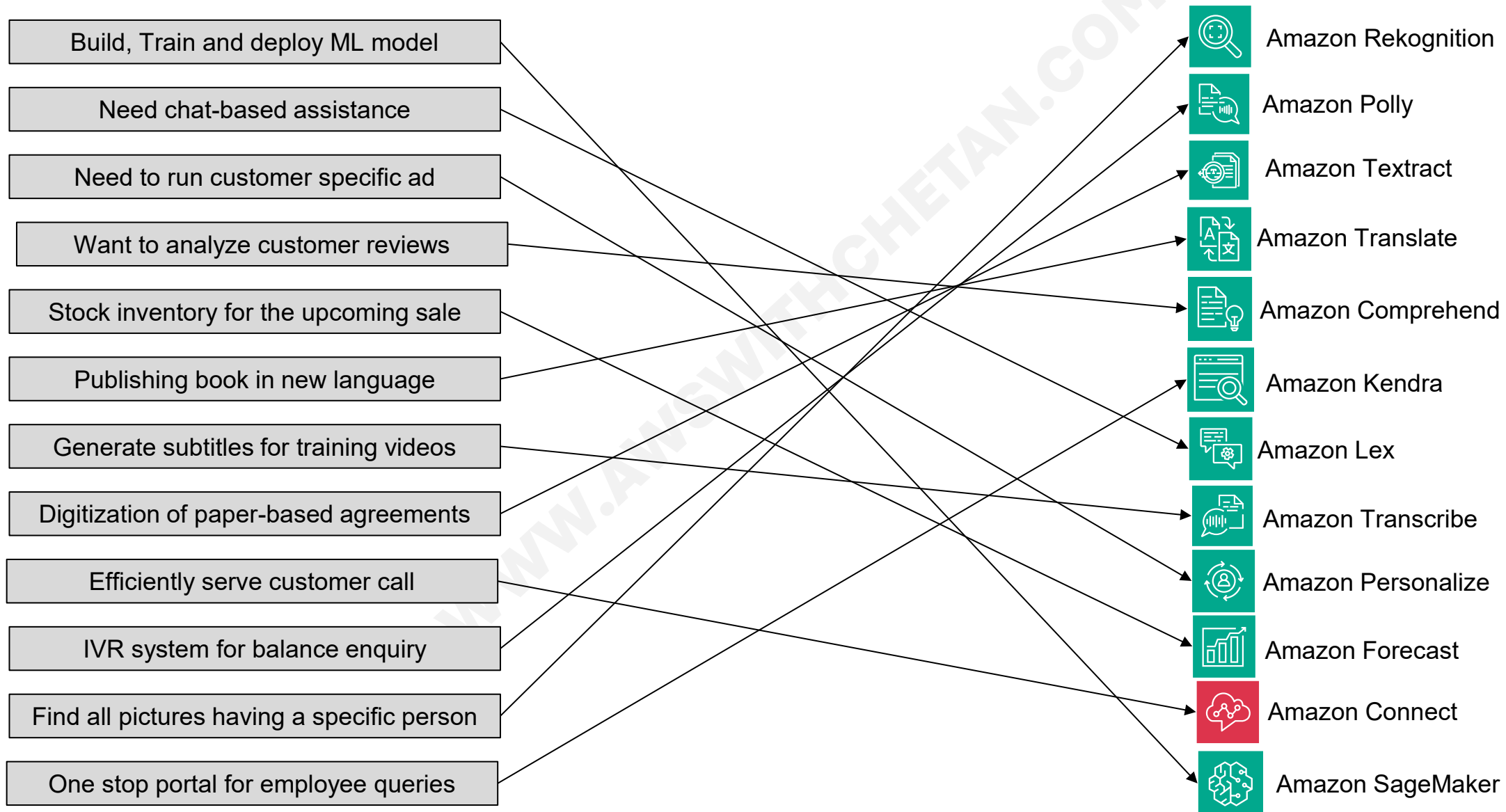


AWS AI and ML services

- Amazon Rekognition – Computer vision-based object detection (Face, celebrity, text, scene) in Images and Videos
- Amazon Transcribe - Speech to text
- Amazon Polly - Text to Speech
- Amazon Textract – Extract text from scanned documents
- Amazon Translate – Translation from one language to other
- Amazon Comprehend – Entity extraction, sentiment analysis
- Amazon Kendra – Enterprise search engine for employees and customers
- Amazon Lex - Build conversational chatbots
- Amazon Connect - Cloud contact center
- Amazon Personalize – Personalized recommendation engine
- Amazon Forecast - Build highly accurate forecasts using ML models used by Amazon.com
- Amazon Fraud Detector – Detect potential fraud in transactions, payments
- Amazon SageMaker - Machine learning platform for developers and data scientists



AWS AI and ML services - When to use what?

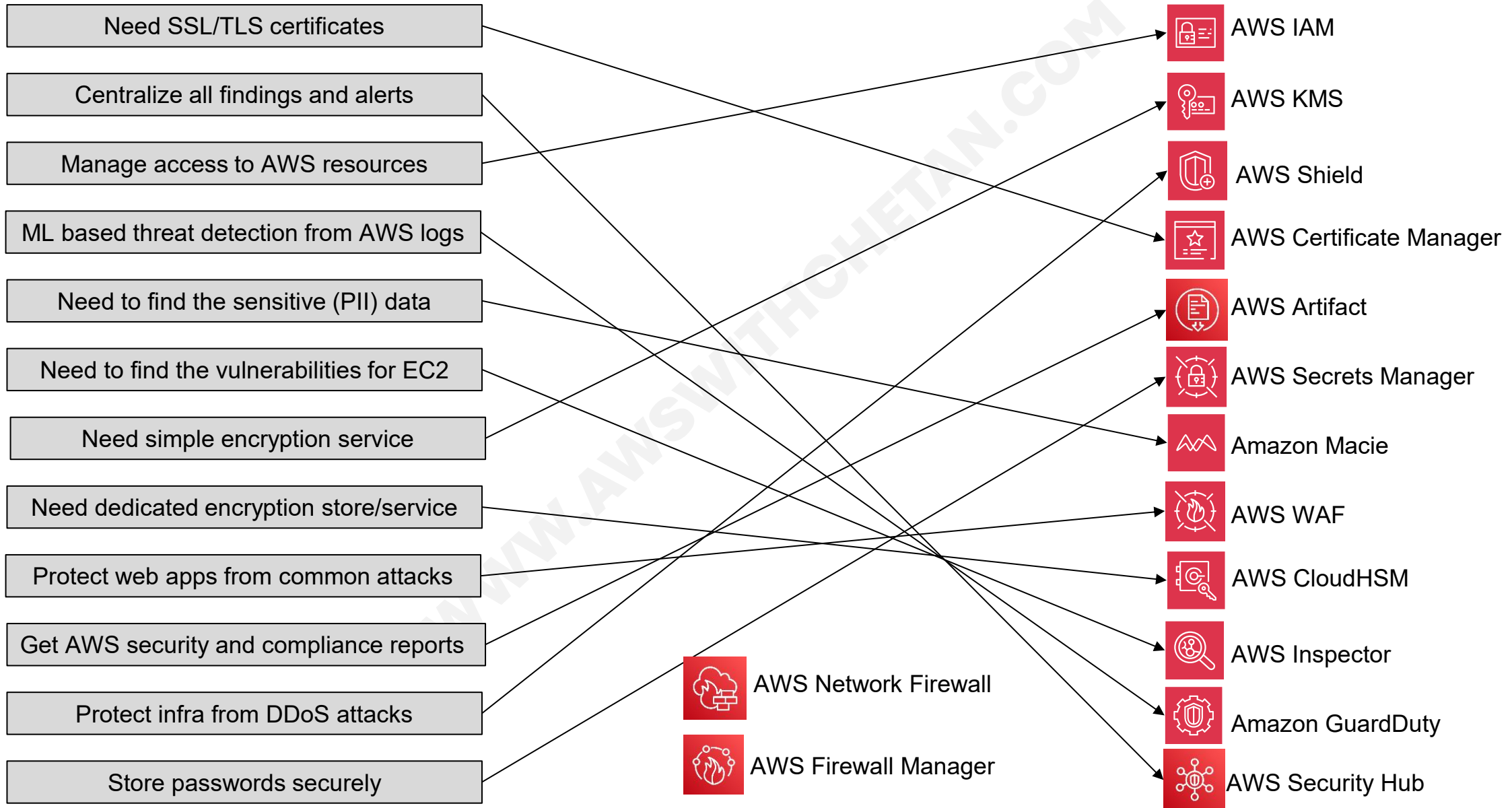


AWS Security Services

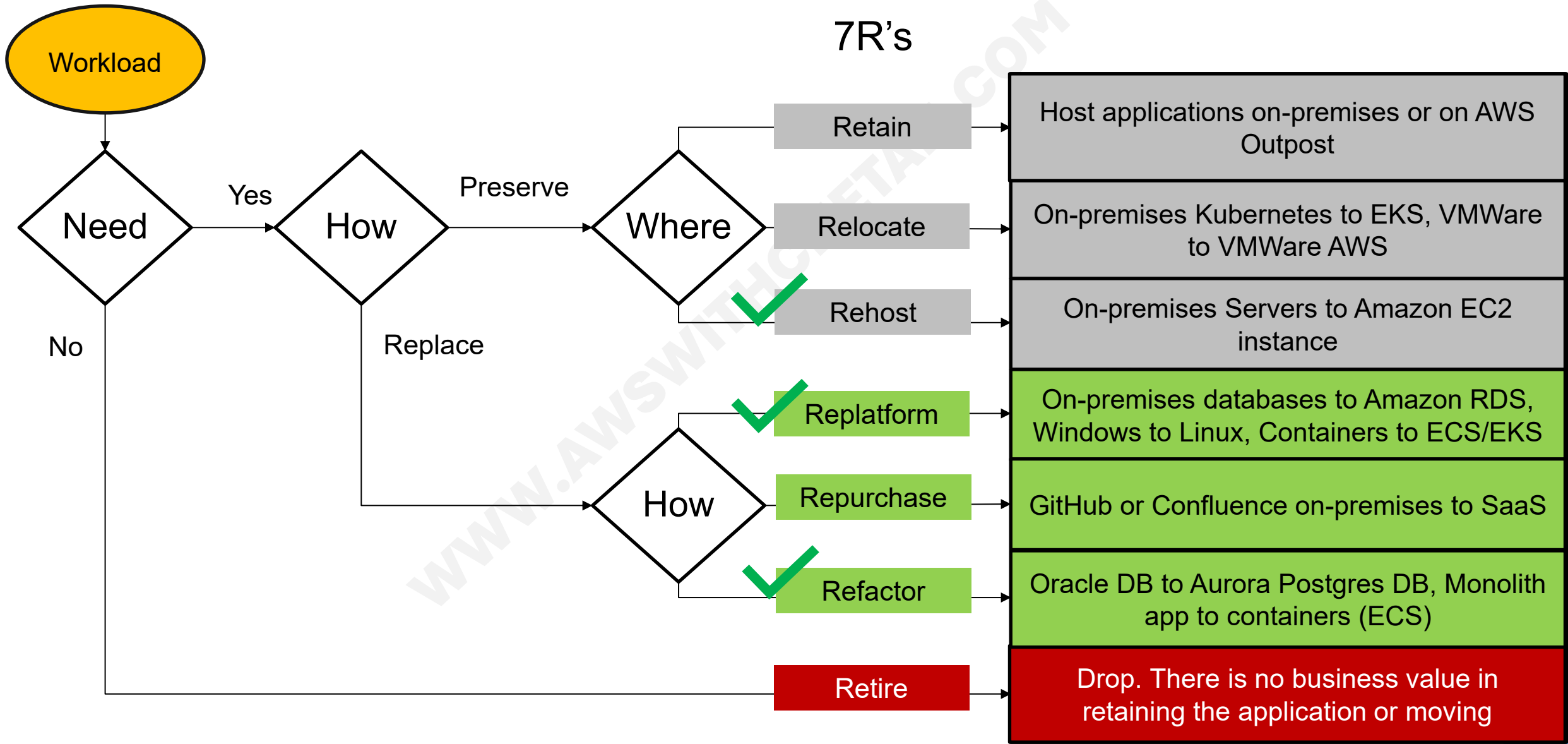
- Amazon KMS - Create and control keys used to encrypt or sign your data
- AWS CloudHSM - Manage single-tenant hardware security modules (HSMs) on AWS
- AWS Certificate Manager - Provision and manage SSL/TLS certificates with AWS services and connected resources
- AWS Secrets Manager - Centrally manage the lifecycle of secrets
- AWS Macie - Discover and protect your sensitive data stored in S3.
- AWS WAF - Protects your web applications from common exploits.
- AWS Shield - Maximize application availability and responsiveness with managed DDoS protection.
- Amazon Inspector - Find software vulnerabilities in EC2, ECR Images, and Lambda functions
- Amazon GuardDuty - Protect AWS accounts with intelligent threat detection by analyzing AWS services logs
- AWS Security Hub - Centrally gather security findings and security alerts from multiple AWS accounts.
- Amazon Detective - Find the root cause of security issues or suspicious activities
- AWS Network Firewall - A stateful intrusion detection and prevention service
- AWS Firewall Manager - Centrally configure and manager Firewall rules across AWS accounts
- AWS Artifact - A portal to access and download security and compliance reports and agreements from AWS



AWS security services - When to use what?



Migration 7R strategy



AWS Migration Services

- Application Discovery Service (ADS) - Capture system and applications inventory and dependencies
- AWS Migration Hub - Consolidate discovery data across regions. Track the entire migration centrally.

AWS Application Migration Service (MGN)

- Lift-and-shift (rehost) migration service for servers to AWS
- Performs continuous block-level replication of server disks
- Maintains near-zero data loss (low RPO) and No downtime during replication phase
- Primary use case: Replicate application and databases to Amazon EC2

AWS Database Migration Service (DMS)

- Migrates on-premises databases to Amazon RDS, Aurora, Amazon Redshift or DynamoDB databases
- Supports both homogeneous database migration (like to like) and heterogeneous migration (e.g. SQL server to Aurora Postgres) using SCT (Schema Conversion Tool)



AWS Data Migration and Hybrid storage services

AWS DataSync

- Moves data between on premises and AWS Storage services
- Supports NFS, SMB, HDFS storage and AWS storage services such as S3, EFS and FSx.
- Supports features like scheduling, bandwidth limit, parallel tasks, transfer failure handling etc.

AWS Storage Gateway

- Connects on-premises storage to AWS cloud storage services enabling hybrid cloud storage for backup, archiving, and disaster recovery.
- There are following different storage gateway types:
 - Amazon S3 File Gateway - Access to S3 (Data lake, archive images/videos, database backups to S3)
 - Amazon FSx File Gateway – Access to FSx for Windows Server (File backups)
 - Tape Gateway - Replace physical tapes with virtual tapes in AWS (Tape backup)
 - Volume Gateway – Access to EBS volumes (On-premises cache, backup)



Disaster Recovery and Backup

DR Objectives:

- Recovery point objective (RPO): The maximum acceptable amount of time since the last data recovery point.
- Recovery time objective (RTO): The maximum acceptable delay between the interruption and restoration of service.

DR Strategies

- Active-Active - Highest availability, Near-zero downtime, Highest cost (as multiple sites running)
- Active-Passive - Passive Site does not serve traffic in normal conditions. In case of failure in the Primary site, the passive site is brought up and traffic is shifted.
- Active-Passive DR Strategies:
 - Backup and Restore - Take regular backups and move to passive site. High recovery time. Most cost effective.
 - Pilot Lite - Continuous data replication to passive site. Lesser recovery time but higher cost.
 - Warm StandBy - Infrastructure running at lower capacity in passive site, scales up in case of failover. Less recovery time but higher cost compared to pilot lite.

AWS Elastic Disaster Recovery (DRS) - Enables fast, reliable recovery of on-premises and cloud-based applications

AWS Backup - Centralizes and automates backups for data across AWS services and on-premises.



AWS Shared Responsibility Model

- Security and Compliance is a shared responsibility between AWS and the customer.
- AWS operates, manages and controls the physical security of the facilities in which the service operates.
- The customer assumes responsibility and management of the guest operating system, application software, data, firewalls (e.g. Security groups), IAM permissions etc.
- Customer Responsibility - EC2 guest operating system patching, Security group rules, Software installed in the EC2, SSH access to EC2, IAM role for EC2, S3 bucket policy & permissions, Enabling data encryption, RDS database data, user creation and permissions, Manual backups, VPC Public and Private network configurations etc.
- AWS Responsibility - Securing the physical infrastructure, including data centers, servers, networking hardware. EC2 underlying physical host, RDS automated backups, RDS maintenance & patching, S3 data multiple copies across AZs, AWS lambda, fargate runtime platform environment security etc.
- **AWS Acceptable Use Policy** governs your use of the services offered by Amazon Web Services e.g. not allowed to use AWS services for any illegal or fraudulent activities, violating rights of others, spamming etc.



AWS Account Management

AWS Organization

- AWS organization consists of a management account and zero or more member accounts
- Member accounts can be organized into a group called Organization Units for budgetary, security and compliance needs (up to 5 levels of nesting).
- Consolidated Billing - Single bill for all member accounts (to be paid by Management account)
- Restrict AWS account/users/role access using Service Control Policies (SCP)
 - SCP governs the maximum available permissions for the IAM users and roles in the AWS organization
 - SCPs do not grant additional permissions
 - Applied at the OU level or Member Account level

AWS Control Tower

- Deploy a multi-account AWS environment with pre-configured best practices (Landing Zone)
- Supports features like Account Factory, Dashboard

AWS Resource Access Manager (RAM)

- Share AWS resources with other AWS accounts inside or outside of AWS organization.
- Supports sharing VPC subnets, Transit gateway, Route53, EC2 dedicated host, CloudHSM and more



AWS Billing and Cost Management

- AWS Pricing calculator – Create cost estimates and share with others (free tool)
- AWS Bills – Overall monthly bills with service level breakup
- Cost Explorer – Detailed cost breakup and analysis and also cost forecast
- Cost Anomaly detector – Detect unusual spend using historic usage data and machine learning
- Data Export – Raw dataset for created your own analysis and dashboard (Formerly CUR)
- AWS Billing Alert – CloudWatch alarm in us-east-1 (N. Virginia) region to alert when AWS usage exceeds
- AWS Budgets – Create budgets for overall AWS usage or service usage or RI/Savings plan usage
- Cost Allocation Tags – Group the AWS resources by resource tags and create cost categories
- Cost Optimization Hub – Recommendation for purchasing Savings Plan, Reserved Instances
- AWS Cost Optimizer – Recommendations to optimize cost by right sizing resources (e.g. EC2 size)
- AWS Free tier dashboard – 12 months free services, Free trials and Always free services



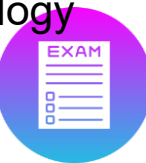
AWS Well-Architected Framework

AWS Well-Architected Six Pillars

1. **Operational excellence** - The ability to support development and run workloads effectively, gain insight into their operations, and to continuously improve.
2. **Security** - The ability to protect information, systems, and assets while delivering business value
3. **Reliability** - The ability of a system to recover from infrastructure or service failures, dynamically acquire computing resources to meet demand, and mitigate disruptions
4. **Performance efficiency** - The ability to use computing resources efficiently to meet system requirements, and to maintain that efficiency as demand changes and technologies evolve.
5. **Cost Optimization** - The ability to avoid or eliminate unneeded cost or suboptimal resources
6. **Sustainability** - Guidance on how AWS can help you reduce your carbon footprint, and best practices you can use to improve the sustainability of your workloads.

AWS Well-Architected Lens

- AWS Well-Architected Lenses extend the guidance offered by AWS Well-Architected to specific industry and technology domains, such as Machine Learning (ML), Data Analytics, Serverless, IoT, SAP, Financial services and more



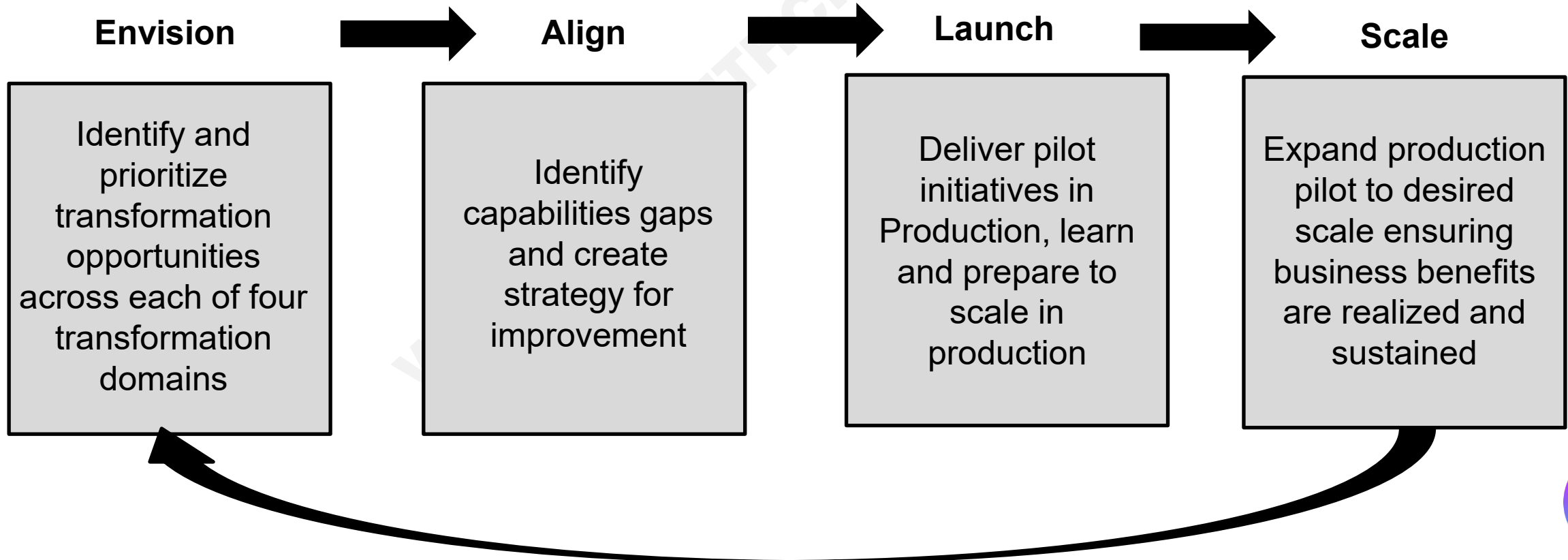
AWS Cloud Adoption Framework (CAF)

- **Four Transformation Domains** - Technology, Process, Organization, Products
- **Six foundational capabilities and perspective** - Business, People, Governance, Platform, Security, and Operations
- **Business** - Management of Strategy, Portfolio, Innovation, Product, Partnership, Data Monetization etc.
 - Key Stakeholders - CEO, CFO, COO, CIO, CTO
- **People** - Culture evolution, Leadership, Cloud Fluency, Workforce transformation, Org. alignment etc.
 - Key Stakeholders - CIO, COO, CTO, Cloud Director, and Cross-functional and enterprise-wide leaders
- **Governance** - Program and Project management, Risk management, Cloud Financial, Data governance, Data curation
 - Key Stakeholders - Chief Transformation Officer, CIO, CTO, CFO, CDO, and Chief Risk Officer (CRO)
- **Platform** - Data Architecture, platform engineering, data engineering, application development, DevOps
 - Key Stakeholders - CTO, technology leaders, architects, and engineers
- **Security** - Security assurance, IAM, Threat detection, Vulnerability, Data protection, Incident response, App security
 - Key Stakeholders - CISO, Audit leaders, and security architects and engineers
- **Operations** - Observability, Event handling, Incident management, Change and release management, performance and capacity management, Configuration management, Patch management, Availability & Continuity
 - Key Stakeholders - Infra and Ops leaders, Site Reliability Engineers (SREs), IT service managers



AWS CAF – Four cloud transformation phases

- The AWS CAF recommends four iterative and incremental cloud transformation phases shown in the following figure.



AWS Support plans (until end of 2026)

What to Compare	Basic	Developer	Enterprise On-Ramp	Enterprise
Best for	All AWS customers	Development & testing workloads	Production workloads	Business / mission-critical workloads
Technical Support	No technical support	Business-hours email	24×7 phone/web/chat	24×7 phone/web/chat
Response Time	—	General: <24 business hrs	Business-critical: <30 min	Business-critical: <15 min
Technical Account Manager (TAM) support			✅ TAM access	✅ Designated TAM
Trusted Advisor	Limited checks	7 core checks	Full set of checks	Full set + Trusted Advisor Priority
Architecture Guidance		General guidance	Proactive & consultative	Proactive & strategic
Take away	Included FREE	Dev/Test	Production + TAM	Critical + TAM + fastest response

Note: Developer Support & Enterprise On-Ramp are being discontinued (Jan 1, 2027) and replaced by AWS Business Support+ for customers transitioning from these plans. Enterprise Support continues.



AWS Trusted Advisor

Trusted Advisor analyzes your AWS environment and provides recommendations across 6 categories:

1. Cost Optimization

- Reduce unnecessary costs → Idle/underutilized EC2, unattached EBS volumes

2. Performance

- Improve speed & responsiveness → EC2 performance, EBS configuration

3. Security

- Identify security gaps → Open security groups, S3 permissions, root MFA

4. Fault Tolerance

- Improve resilience & redundancy → RDS Multi-AZ, resource redundancy

5. Service Limits

- Monitor service quotas → EC2 quota approaching its limit

6. Operational Excellence

- Improve operational practices → Logging, monitoring, recommended configurations



AWS Partner Network (APN), Professional Services, Community

AWS Partner Network (APN)

- Independent organizations that provide AWS consulting, migration, implementation, and managed services. There are 3 types of the partners - Consulting Partners, Technology Partners and Training Partners
- Consulting Partners - Helps using AWS (IT Consulting, Solution Design, Application Development, Managed Services)
- Technology Partners - Builds tech that runs on AWS - Database providers, Network/DevOps tools providers, SAP etc.

AWS Professional Services (AWS ProServe)

- AWS-employed experts who help customers design, migrate, implement, and optimize AWS solutions.

AWS IQ - Marketplace that connects customers with AWS-certified experts for on-demand help

AWS Managed Services (AMS)

- AWS service that helps organizations operate AWS environments at scale, including monitoring, incident management, security, and operational support.

AWS re:post

- AWS re:post is a public forum to discuss AWS issues. There is also private version of re:post which you can use inside your organization to collaborate internally.



Other AWS Services - good to know

- AWS Marketplace – A marketplace for ISVs to sell their products on AWS
- AWS License Manager – Centrally manage the software licenses across AWS and on-premises
- AWS End User Computing Services
- Amazon Workspaces – Managed VDI service
- Amazon Appstream – Managed Application streaming service
- AWS Backup – Automate backups across AWS and on-premises
- AWS Step function – Build workflows for distributed applications and microservices



That's it ..

All the best for your exam !

After you clear your exam, please share about your learning experience as a feedback for this course and I hope to receive 5-star rating from you :)



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Thank you